Summary in Graph

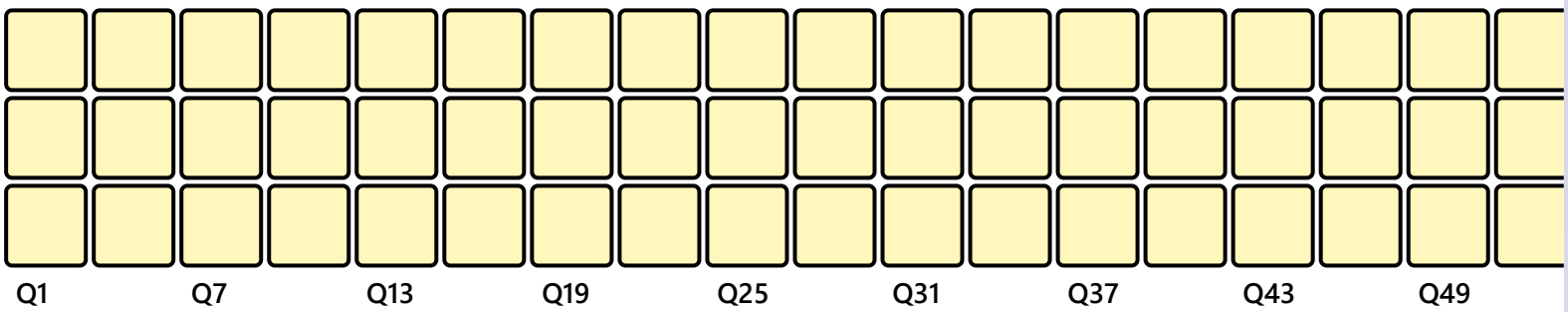
Exam Summary (GATE Overflow CS Test Series 2026 | Mock GATE 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	65 (30 + 35)
Total Marks:	100 (30 + 70)
Exam Duration:	180 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (14) out of 144 users

EXAM RESPONSE EXAM STATS FEEDBACK

Aptitude

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Verbal Aptitude

It is a standard practice _____ a civil servant to sign checks issued by the Treasury Department to ensure that government payments are not biased.

A. for
B. to
C. on

D. upon

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Verbal Aptitude

Which underlined part of the below sentence has an error?

Only lately $\underbrace{\text{did the chief officer realized that the}}_{P}$ $\underbrace{\text{findings provided to him}}_{Q}$ $\underbrace{\text{by his employee were false}}_{R}$.

A. P
B. Q
C. R
D. No error

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Verbal Aptitude

The questions given below consists of a pair of related words followed by another pair of related words. Select the option that best expresses the relation same as the original pair.

Heavy : Light :: Smooth : _____

A. Hard
B. Dark
C. Rough
D. Frail

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Analytical Aptitude

Find the missing group of letters in the series:

ED, JIH, QPON, _____

A. $YWXVU$
B. $AZYXW$
C. $ZYXWV$
D. $BAZYX$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

If a is a positive odd number and b is a negative even number, then which of the following must be a positive even number?

A. $a - b$
B. ab

C. $-ab$
D. $a+b$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Numerical Type | Award: 2 | Penalty: 0 | Spatial Aptitude

If the number of edges in Cube, Tetrahedron, Sphere, and Triangular prism are α, β, γ, and ρ, respectively. Then the value of $\alpha+\beta-\gamma+\rho=$_____

Your Answer: | Correct Answer: 27 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Numerical Type | Award: 2 | Penalty: 0 | Spatial Aptitude

If the sum of number of vertices and faces in a polyhedron is 2021, then the number of edges in the polyhedron is _____

Your Answer: | Correct Answer: 2019 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Analytical Aptitude

In each of the questions below, two statements are given followed by conclusions: I, II, and III. You have to take the given statements to be true even if they seem to be at variance from commonly known facts. Read the conclusions and decide which of the given conclusions logically follows from the given statements disregarding commonly known facts.

Statements:

1. Only a few coaches are players.
2. No coaches are fielders.

Conclusions:

I. Some fielders are definitely not players.
II. All players being fielders is a possibility.
III. All coaches can never be players.

A. Only II and III follow
B. All of I, II, and III follow
C. Only I follows
D. Only III

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Verbal Aptitude

Most bicycle helmets provide good protection for the top and back of the head, but little or no protection for the temple regions on the sides of the head. A study of head injuries resulting from bicycle accidents showed that a large proportion were caused by blows to the temple area. Therefore, if bicycle helmets protected this area, the risk of serious head injury in bicycle accidents would be greatly reduced,especially since _____ Which of the following most logically completes the passage?

A. the bone in the temple area is relatively thin and impacts in that area are thus very likely to cause brain injury.

B. among the bicyclists included in the study's sample of head injuries, only a very small proportion had been wearing a helmet at the time of their accident.
C. bicyclists generally land on their arm or shoulder when they fall to the side, which reduces the likelihood of severe impacts on the side of the head.
D. a helmet that included protection for the temples would have to be somewhat larger and heavier than current helmets.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 03sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Spatial Aptitude

The area of the closed region by the equation $|x| + |y| = 4$ in the two - dimensional plane is?

A. 32π
B. 16
C. 8
D. 32

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Technical

Q #1 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

Which of the following is/are necessary conditions for deadlock? (Mark all the appropriate choices)

A. Mutual exclusion
B. Hold and wait
C. Preemption
D. Circular wait

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Assuming $i = \sqrt{-1}$ and t is a real number ,

$$I = \int_0^{\frac{\pi}{3}} e^{it} dt$$

A. $\frac{\sqrt{3}}{2} + i\frac{1}{2}$
B. $\frac{\sqrt{3}}{2} - i\frac{1}{2}$
C. $\frac{1}{2} + i\frac{\sqrt{3}}{2}$
D. $\frac{1}{2} + \left(1 - \frac{\sqrt{3}}{2}\right)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

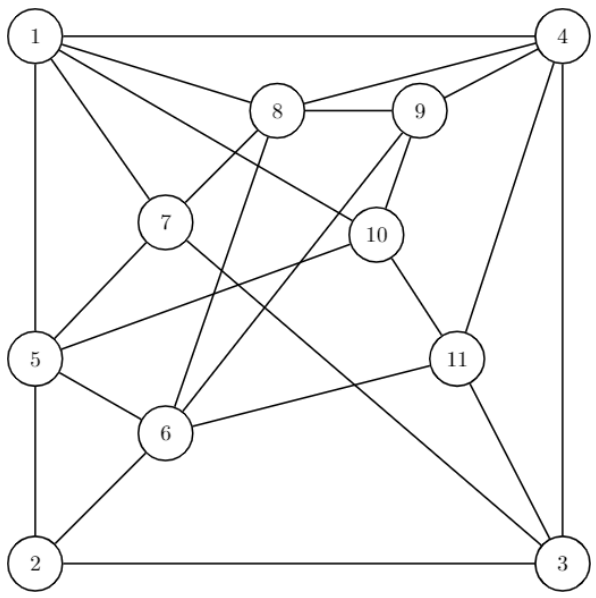
Which of the following commands are not from the File Transfer Protocol (FTP)?

A. PORT
B. PROMPT
C. HELO
D. RECV

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Graph Theory

What is the minimum number of colors required to color the below graph?



A. 2
B. 3
C. 4
D. 5

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Numerical Type | Award: 1 | Penalty: 0 | Operating System

A single level paging scheme uses a Translation Look-aside Buffer (TLB). A TLB-access takes 20 nanoseconds and the main memory access takes 70 nanoseconds. If the TLB hit ratio is 90% and there are no page-faults, the effective memory access time (in nanoseconds) is _____

Your Answer: | Correct Answer: 97 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

The Boolean multiplication symbolizes which of the following function?

A. Logical XOR function
B. Logical OR function
C. Logical AND function
D. None of the above

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

A perfect binary tree is a binary tree in which all interior nodes have two children and all leaves have the same depth or same level. Let T be a full binary tree with height 'h' containing $2^{(h+1)} - 1$ nodes. Which one of the following is the value of the sum of height of the nodes of the tree T (assume height of a leaf node as 0)?

A. $2^{(h+1)} - 1$
B. $(h-1) \times 2^{(h+1)} + 2$
C. $2^{(h+1)} - 1 - h$
D. $2^{(h+1)} - 2 - h$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

The preorder traversal sequence of a binary search tree is

$$26, 21, 11, 6, 2, 9, 13, 18, 23, 38, 30, 29, 42, 41, 50, 43, 60.$$

Which one of the following is the postorder traversal of the same tree?

A. $2, 9, 6, 18, 13, 11, 21, 23, 41, 29, 30, 43, 60, 50, 42, 38, 26$
B. $26, 38, 42, 50, 60, 41, 43, 21, 29, 30, 23, 11, 13, 18, 6, 2, 9$
C. $26, 38, 42, 50, 60, 43, 41, 30, 29, 21, 23, 11, 13, 18, 6, 9, 2$
D. $2, 9, 6, 18, 13, 11, 23, 21, 29, 30, 41, 43, 60, 50, 42, 38, 26$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 Multiple Select Type | Award: 1 | Penalty: 0 | Computer Networks

Which of the following is/are NOT correct? (Mark all the appropriate choices)

A. In circuit switching, the resources need to be reserved during the setup phase
B. In a packet-switched network, there is no resource reservation; resources are allocated on demand
C. In packet-switching, all packets belonging to the same source and destination travel the same path
D. In virtual-circuit switching, all packets belonging to the same source and destination travel the same path

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 Numerical Type | Award: 1 | Penalty: 0 | Operating System

Consider processes $P_i, i = 1, \ldots, 20$ coded as follows:

```
    repeat
    V(mutex)
    {Critical section}
    P(mutex)
5.  forever
```

Here, mutex is a binary semaphore initialized to 0. The code for P_{21} is identical except it uses $\mathrm{P}\ (\mathrm{mutex})$ in place of $\mathrm{V}\ (\mathrm{mutex})$ and vice versa. What is the largest number of processes that can be inside the critical section at any moment _____

Your Answer: | Correct Answer: 21 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Select Type | Award: 1 | Penalty: 0 | Digital Logic

Let A, B, C be three sets. $\oplus$ denotes the symmetric difference operation.

Which of the following is/are true? (Mark all the appropriate choices)

A. $(A \oplus B) \oplus C = A \oplus (B \oplus C)$
B. $A \oplus B = B \oplus A$
C. If $A \oplus B = A \oplus C$ then $B = C$
D. $A \cap (B \oplus C) = (A \cap B) \oplus (A \cap C)$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Numerical Type | Award: 1 | Penalty: 0 | Operating System

Let the page fault service time be 25 milliseconds(ms) in a computer with average memory access time being 40 nanoseconds (ns). If one page fault is generated every $100,000$ memory accesses, the effective access time in nanoseconds for memory is _____

Your Answer: | Correct Answer: 289 : 290 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

With 64 bit virtual addresses, a 4 KB page and 16 GB of RAM, an inverted page table requires _____

A. 2 million entries
B. 4 million entries
C. 512 thousand entries
D. 1 million entries

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Set Theory & Algebra

Let R and S be relations on a set A. Assuming A has at least three elements, which of the following statement is/are TRUE?

- Statement 1 : If R and S are symmetric then $R \cap S$ is symmetric.
- Statement 2 : If R and S are symmetric then $R \cup S$ is symmetric.

A. Both statements are TRUE
B. Statement 1 is TRUE but Statement 2 is FALSE
C. Both statements are FALSE
D. Statement 2 is TRUE but Statement 1 is FALSE

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Combinatory

The minimum number of integers to be selected from the set $S = \{1, 2, \ldots, 9\}$ so that the difference of at least two of the integers is guaranteed to be 5 is _____

A. 4
B. 5
C. 6
D. 7

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Combinatory

The number of ways 12 students in a class can take 3 subject tests if each test is taken by 4 students and each student take exactly one test is ____

A. 34000
B. 34750
C. 34650
D. 35000

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Numerical Type | Award: 1 | Penalty: 0 | Mathematical Logic

Suppose of all the possible ways to choose two representative among 12 players in a team, there are exactly 45 different ways to choose one or more female as a representative. The number of different males in the team is _____

Your Answer: | Correct Answer: 7 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Combinatory

A Soldier was appointed for planning a parade layout. His boss than informed him that he was planning to add more participants. "There is no way I can finish this!",Soldier said." There must be more than a billion possibilities!".
The minimum number of participants that will give more than a billion possible orderings is ____

A. 12
B. 13
C. 14
D. 15

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Combinatory

Which one of the following is a closed form expression for the generating function of the sequence $\{a_n\}$ where $a_n = \binom{n+4}{n}$ for $n = 0, 1, 2, \ldots$?

A. $\frac{1}{(1-x)^5}$
B. $\frac{5}{(1-x)}$

C. $\frac{1}{(1-x)^4}$
D. $\frac{x}{(1-x)^5}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 Multiple Select Type | Award: 1 | Penalty: 0 | Mathematical Logic

Which of the following is/are well formed propositional formulae? (Mark all the appropriate choices)

A. $\vee pq$
B. $(\neg(p \rightarrow q \wedge p))$
C. $(\neg(p \rightarrow (q = p)))$
D. p

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 Multiple Select Type | Award: 1 | Penalty: 0 | Mathematical Logic

Which of the following is the correct first order logic representation for the below sentence?

$$\text{“Bill has at most one sister”}$$

here $\text{SisterOf}(x, y) = x$ is sister of y (Mark all the appropriate choices)

A. $\forall x \forall y(\text{SisterOf(x,Bill)} \wedge \text{SisterOf(y,Bill)} \rightarrow (x = y))$
B. $\forall x \neg \text{SisterOf(x,Bill)} \vee \exists x(\text{SisterOf(x,Bill)} \wedge \forall y(\text{SisterOf(y,Bill)} \rightarrow (x = y))$
C. $\forall x \text{SisterOf(x,Bill)} \vee \exists x(\text{SisterOf(x,Bill)} \wedge \forall y(\text{SisterOf(y,Bill)} \rightarrow (x = y))$
D. $\exists x(\text{SisterOf(x,Bill)} \wedge \forall y(\text{SisterOf(y,Bill)} \rightarrow (x = y))$

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 Numerical Type | Award: 1 | Penalty: 0 | Set Theory & Algebra

The Poset $(D_{210}, /)$ is a boolean algebra and it's hasse diagram is isomorphic to the Poset $(P(S), \subseteq)$ Hasse diagram where S is finite set. The cardinality of set S is_____ (D_n is set of all divisors).

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

Consider the following C program run on a system using IEEE-754 representation for floating-point numbers.

```c
#include<stdio.h>
int main()
{
  float a=2.4,b=2.5;
  if(a==2.4)
  {
    printf("2021");
  }
  if(b==2.5)
  {
    printf("2020");
  }
  return 0;
}
```

What will be the output of the above program?

A. 20212020
B. 2020
C. 2021
D. No output

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

A station transmits frames of length 200 bits at the rate of 2 Mbps to another station 2 kilometers apart. The velocity of propagation is 3×10^8 m/s. Assuming that each station generates frames at an average rate of 10000 frames per second, what is the probability of collision for pure ALOHA?

A. 0.53
B. 0.68
C. 0.73
D. 0.88

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

Consider the following C Code

```c
#include<stdio.h>
int main()
{
   int i=0;
   do
   {
       do
       {

          while(i++<2);

       }while(i++<4);

   }while(i++<=9);

   printf("%d",i++);

   return 0;
}

```

What will be the output of the above program?

A. 13
B. 11
C. 12
D. 10

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

Two nodes, A and B, communicate through a store and forward network. Node A is connected to the network by a 100 Mbps link, while node B is connected by a 500 Mbps link. Node A sends two back-to-back packets of 1000 bits each. The difference between the arrival times of the two packets at B is 1 ms. What is the smallest capacity of a link along the path between A and B (in Mbps)?
Note: Assume that there are no other packets in the network except the ones sent by A, and ignore the packet processing time. Assume both packets follow the same path, and they are not reordered. The arrival time of a packet at a node is defined as the time when the last bit of the packet has arrived at that node.

A. 1
B. 2
C. 5
D. 10

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

Consider an infinite queue that can send data at 10 Kbps. Assume the following arrival traffic: During every odd second the queue receives an 1000 bit packet every 50 ms. During every even second the queue receives no data. Assume an interval $'I'$ of 10 sec starting with an odd second (i.e., a second in which the queue receives data). At the beginning of interval I the queue is empty. What is the maximum queue size (i.e no of packets) during interval $'I'$?

A. 10
B. 11
C. 12
D. 13

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

Which of the following statement is TRUE regarding the way CSMA/CD react to collisions?

A. All systems jam the network, and then all begin transmitting again
B. Hosts involved in a collision send an RTS signal indicating a time frame in which to retransmit
C. Hosts involved in the collision send a jam signal, and then run an algorithm before retransmitting
D. Collisions do not occur on CSMA/CD

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

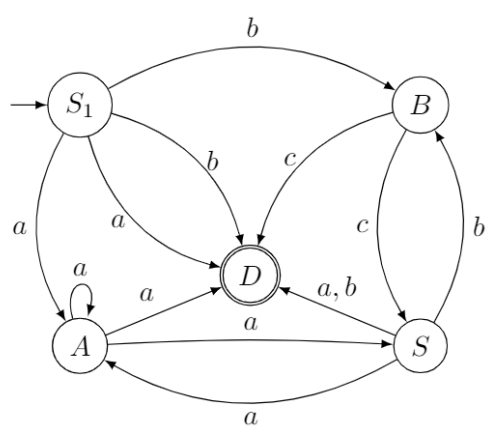
Alice is fond of doing programming, she usually takes part in different coding contest and mostly get gold medal because of the efficiency of her codes. Out of curiosity to know whether Alice is as that much capable in programming or not his sister has given her a program. She said "There are 2 matrices P and Q both of order 5×5, P is lower triangular matrix and Q is upper triangular matrix, you have to find the product of P and Q i.e., PQ and print the minimum number of algebraic operations used, but mind it, the program has to be as efficient as possible". What will be the value printed by Alice's program?

A. 225
B. 110
C. 85
D. 55

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Which of the following is the regular grammar for the following finite automata? (Mark all the appropriate choices)



A. $S \rightarrow a \mid aA \mid bB \mid \varepsilon$
$A \rightarrow aA \mid aS$
$B \rightarrow cS \mid \varepsilon$
B. $S \rightarrow a \mid b \mid aA \mid bB$
$A \rightarrow a \mid aA \mid aS$
$B \rightarrow c \mid cS \mid \varepsilon$
C. $S \rightarrow a \mid aA \mid bB$
$A \rightarrow aA \mid aS$
$B \rightarrow cS \mid \varepsilon$

D. $S \rightarrow a \mid b \mid aA \mid bB$
$A \rightarrow a \mid aA \mid aS$
$B \rightarrow a \mid cS \mid \varepsilon$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #31 Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Which of the given strings is/are NOT generated by the below context-free grammar? S is the start symbol and the set of non-terminal symbols is $\{a, b\}$.

The production rules are:

- $S \rightarrow bS \mid aAaS \mid \varepsilon$
- $A \rightarrow bA \mid \varepsilon$

A. $ababbbbb$
B. $aaaaaaaaa$
C. $bbbbabbbb$
D. $aaaabaaaa$

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #32 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C

What will be the output of the following C program?

```c
#include<stdio.h>
void foo(int a, int *p1, int **p2)
{
  static int b;
  *p1 = 6;
  *p2 = &b;
}

int main()
{
  int a, *p1 = &a;
  a = 9;
  foo(a, p1, &p1);
  printf("%d,%d",a, *p1);
}
```

A. $6, 0$
B. $9, 6$
C. $6, 6$
D. Garbage Value

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #33 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

Consider the virtual page reference string

$$1, 2, 3, 4, 2, 1, 6, 5, 1, 2, 2, 3$$

on a demand paged virtual memory system running on a computer system that has main memory size of 3 page frames which are initially empty. Let LRU, FIFO and OPTIMAL denote the number of page faults under the corresponding page replacement policy. Then

A. $\text{OPTIMAL} < \text{LRU} < \text{FIFO}$
B. $\text{OPTIMAL} < \text{FIFO} = \text{LRU}$
C. $\text{OPTIMAL} = \text{LRU}$
D. $\text{OPTIMAL} = \text{FIFO}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #34

Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Which of the following languages is/are recursively enumerable but not recursive? Here, $\langle M \rangle$ denote an encoding of Turing machine M. (Mark all the appropriate choices)

A. $L = \{\langle M \rangle \mid 0 \in L(M)\}$
B. $L = \{\langle M \rangle \mid L(M) \text{ is finite}\}$
C. $L = \{\langle M \rangle \mid L(M) \text{ is recursive}\}$
D. $L = \{\langle M \rangle \mid L(M) \text{ is recursively enumerable}\}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #35

Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

An inversion is a pair of numbers at indices i and j, such that $i < j$, number$[i] >$ number$[j]$. Consider the following array $A : A[\,] = \{1, -3, -1, -2, 5, 4, 0\}$
Which of the following is the sorted sequence of the number of inversions of each element of the array? (if the element has no inversion consider inversion count as 0).

A. $0, 0, 0, 1, 1, 1, 5$
B. $0, 0, 1, 2, 2, 3, 4$
C. $0, 0, 0, 1, 1, 2, 4$
D. $0, 1, 1, 2, 2, 3, 4$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #36

Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider the following C function.

```
int func(int n)
{
  int i , j , k , p , q=1;
  for(i=1 ; i<= n ; i++)
  {
    p=0;
    for(k=1 ; k<= n ; k<<=1)
      p++;
    for(j=p ; j>= 1 ; j>>=1);
    q=q<<=1;
  }
 return q;
}
```

Which one of the following is the approximate value returned by the above function?

A. $\Theta(n)$
B. $\Theta(2^n)$
C. $\Theta(2^{(n \log n)})$
D. $\Theta(\log n)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #37 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

The asymptotic growth rate order of the below four functions is given by

- $F_1 = \log(n * \log(n))$
- $F_2 = (\sqrt{\log n})^{\log n}$
- $F_3 = (\log n)!$
- $F_4 = \log(n) * \log(n+1)$

A. $F1 < F4 < F3 < F2$
B. $F4 < F1 < F3 < F2$
C. $F4 < F1 < F2 < F3$
D. $F1 < F4 < F2 < F3$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #38 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider the following C function:

```
float func(int a , int n)
{
   int i,j,k,b=1;
   for(i=0,j=1,k=1;k<n;k++)
   {
      j*=a/k;

      if(k% 2!= 0)
      {
         if(b)
         {
            i=i+j;
            b=0;
         }
         else
         {
            i=i-j;
            b=1;
         }
      }
   }
  return i;
}
```

If the value of n is very large tending to infinity then the returned value approximates.

A. a^b
B. e^a
C. $\sin(a)$

D. $\log(1+a)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #39 Numerical Type | Award: 2 | Penalty: 0 | Operating System

Consider the set of processes whose arrival times and burst times (in milliseconds) are given below:

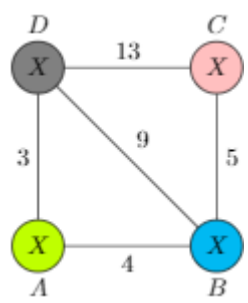
Process Name	Arrival Time	Execution Time
P_1	0	8
P_2	1	4
P_3	2	6
P_4	5	5

Using highest response ratio next scheduling algorithm, the average turn around time in milliseconds for the processes will be _____ (up to two decimal points)

Your Answer: | Correct Answer: 13.25 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #40 Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

There is a network consisting of 4 routers, the weights are mentioned on the edges, weights could be distances or costs or delays. After the DVR(distance vector routing) converses, how many edges which will never be used is _____



Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #41 Numerical Type | Award: 2 | Penalty: 0 | Operating System

Consider a uniprocessor system executing three tasks A, B and C each of which is composed of a sequence of instances as given below:

- A : 3 instances of 2 ms task arriving at intervals of 5 ms, and 1 ms.
- B : 4 instances of 3 ms task arriving at intervals of 2 ms, 1 ms and 3 ms.
- C : 1 instance of 5 ms task.

The priority of each task is the inverse of its period, and the available tasks are scheduled in order of priority, which is the highest priority task scheduled first. Given that all tasks initially arrive at the beginning of the 1st millisecond and task preemptions are allowed, the first instance of C completes its execution at the end of _____ milliseconds.

Your Answer: | Correct Answer: 23 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #42 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

Given four memory partitions of $100\ \text{KB}$, $500\ \text{KB}$, $200\ \text{KB}$, $600\ \text{KB}$ (in order), and memory allocation requests of $300\ \text{KB}$, $496\ \text{KB}$, $280\ \text{KB}$, $90\ \text{KB}$, $50\ \text{KB}$, $80\ \text{KB}$, and $45\ \text{KB}$ (in order), which of the following algorithm(s) can satisfy all the requests?

A. First-fit
B. Best-fit
C. Worst-fit
D. Best-fit and first-fit

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #43 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

Which of the following bits stream in hexadecimal notation correctly represents the maximum possible value represented in IEEE-754 single-precision representation?

A. $FFFFFFFF$
B. $F7FFFFFF$
C. $7FFFFFFF$
D. $7F7FFFFF$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #44 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

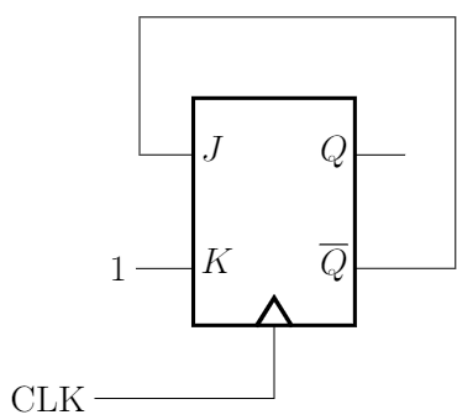
Given a disk-block size of $16\ \text{KB}$ and block-pointer address value of $8\ \text{bytes}$, what is the largest file size that can be accessed using 256 direct addresses and one indirect block?

A. 32 MB
B. 16 MB
C. 20 MB
D. 36 MB

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #45 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

The following JK flip flop circuit is initially reset and then clocked for 5 clock pulses. The output sequence starting from the end of first clock pulse will be ___



A. 01111
B. 10101
C. 10000

D. 01010

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #46 Numerical Type | Award: 2 | Penalty: 0 | Compiler Design

Consider the expression $a + -b * c + d/e$. The minimum number of temporary variables required to create a three-address code for this expression in Static Single Assignment form is _____

Your Answer: | Correct Answer: 5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #47 Numerical Type | Award: 2 | Penalty: 0 | Operating System

Consider FIFO page replacement for the page reference string :

$$0, 1, 2, 3, 0, 1, 4, 0, 1, 2, 3, 4$$

If 3 frames are available, then the number of page faults is a and if 4 frames are available, then the number of page faults is b, the value of $a - b =$_____

Your Answer: | Correct Answer: -1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #48 Numerical Type | Award: 2 | Penalty: 0 | Probability

A school has three divisions-- A, B and C for the tenth grade, having the number of students $40, 45$ and 42 respectively. The pass percentages of students from the three divisions are $96, 97$ and 98 respectively. If a student is chosen at random from the tenth grade and is found to have failed in the exam, what is the probability that it he/she is from the A grade? (Rounded to 2 decimal points)

Your Answer: | Correct Answer: 0.42 : 0.43 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #49 Numerical Type | Award: 2 | Penalty: 0 | Compiler Design

A CFG(Context-Free Grammar) G is given with the following productions where S is the start symbol, A is a non-terminal and a and b are terminals.

- $S \rightarrow aBS \mid bAS \mid \varepsilon$
- $A \rightarrow a \mid bAA$
- $B \rightarrow b \mid aBB$

For the string " $aabaabbb$ " , if the number of steps required to derive the string is $'\alpha'$ and the number of parse trees is $'\beta'$ the the value of $\alpha - \beta$ is _____

Your Answer: | Correct Answer: 8 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #50 Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

In an $IPv4$ datagram, the value of HLEN is 15, the value of total length is 300 and the fragment offset value is 200. The sum of the sequence numbers of the first and the last bytes of the payload is _____

Your Answer: | Correct Answer: 3439 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #51 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider the Sailors-Boats-Reserves $(s-b-r)$ database given below.

s $(\underline{sid}$, sname, rating, age$)$

b $(\underline{bid}$, bname, color$)$

r $(\underline{sid, bid,}$ date$)$

Which of the following describes the colors of boats reserved by Alex?

A. $\Pi_{\text{color}}\left[(\sigma_{\text{sname}=\text{'Alex'}}(s)) \bowtie r\right]$
B. $\Pi_{\text{color}}\left[(\sigma_{\text{sname}=\text{'Alex'}}(s)) \bowtie r \bowtie b\right]$
C. $\Pi_{\text{color}}\left[(\sigma_{\text{sname}=\text{'Alex'}}(s)) \bowtie b\right]$
D. $\Pi_{\text{color}}\left[(\sigma_{\text{sname}=\text{'Alex'}}(b)) \bowtie s \bowtie r\right]$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #52 Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Which of the following is a minimal cover of the set of functional dependencies given:
$\{AB \rightarrow C, C \rightarrow A, BC \rightarrow D, ACD \rightarrow B, D \rightarrow EG, BE \rightarrow C, CG \rightarrow BD, CE \rightarrow AG\}$?
(Mark all the appropriate options)

A. $\{AB \rightarrow C, C \rightarrow A, BC \rightarrow D, ACD \rightarrow B, D \rightarrow E, D \rightarrow G, BE \rightarrow C, CG \rightarrow B, CG \rightarrow D, CE \rightarrow A$
B. $\{AB \rightarrow C, C \rightarrow A, BC \rightarrow D, CD \rightarrow B, D \rightarrow E, D \rightarrow G, BE \rightarrow C, CG \rightarrow D, CE \rightarrow G\}$
C. $\{AB \rightarrow C, C \rightarrow A, BC \rightarrow D, D \rightarrow E, D \rightarrow G, BE \rightarrow C, CG \rightarrow B, CG \rightarrow D, CE \rightarrow G\}$
D. $\{AB \rightarrow C, C \rightarrow A, BC \rightarrow D, D \rightarrow E, D \rightarrow G, BE \rightarrow C, CG \rightarrow B, CE \rightarrow G\}$

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #53 Numerical Type | Award: 2 | Penalty: 0 | Databases

Given relation $R(A, B, C, D, E, F, G, H, I, J)$ and the functional dependencies $\{AB \rightarrow C, BD \rightarrow EF, AD \rightarrow GH, A \rightarrow I, H \rightarrow J\}$, the minimum number of tables required for a decomposition into the $2NF$ is _____

Your Answer: | Correct Answer: 5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #54 Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Consider a 5-stage pipeline - IF (Instruction Fetch), ID (Instruction Decode and register read), EX (Execute), MEM (Memory), and WB (Write Back). All register reads take place in the second phase of a clock cycle and all register writes occur in the first phase. Consider the execution of the following instruction sequence:

- $I_1 : R_1 \leftarrow R_2 * R_3$
- $I_2 : R_3 \leftarrow R_1 + R_2$
- $I_3 : M[R_1 + 200] \leftarrow R_1$
- $I_4 : R_2 \leftarrow R_1 * R_1$

If the number of RAW(Read after write) hazards is denoted by A, WAR(Write after read) hazards by B and WAW(Write after write) hazards by C, then $2A + 3B + 5C =$_____

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #55 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider the following C program:

```c
#include<stdio.h>
int A[100][100],B[100][100];
void Dosomething(int x,int y);
int main()
{
   int a,b;
   scanf("%d%d",&a,&b);
   int i, j;
   for(i = 0; i < a; i++)
   {
       for(j = 0; j < b; j++)
       {
         scanf("%d",&A[i][j]);
       }
   }

   Dosomething(a,b);

   for (i = 0; i < a; i++)
   {
     for (j = 0; j < b; j++)
     {
        printf("%d", B[i][j]);
     }
   }

}

void Dosomething(int a, int b)
{
  int i,j;
  for(i = 0; i < a; i++)
  {
     for(j = 0; j < b; j++)
     {
        B[i][b - 1 - j] = A[i][j];
     }
  }
}
```

What will be the output of the above code if $a = 4$ and $b = 4$ and $A = [[1, 3, 5, 7], [2, 4, 6, 8], [9, 11, 13, 15], [10, 12, 14, 16]]$

A. $1234567891011121314 1516$
B. $1357246891113151012 1416$
C. $1357911131524681012 1416$
D. $7531864215131191614 1210$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

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